

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\bar{\omega}^m \left(\frac{b^{d-1} d^d}{a^d (d-1)^{d-1}} \right) \prod_{i=0}^{r-1} \prod_{h=1}^{d-2} \frac{\Gamma_p(\langle (\frac{h}{d-1} - \frac{m}{q-1}) p^i \rangle)}{\Gamma_p(\langle \frac{h p^i}{d-1} \rangle)} \prod_{\substack{h=1 \\ h \neq \frac{d}{2}}}^{d-1} \frac{\Gamma_p(\langle (\frac{h}{d} + \frac{m}{q-1}) p^i \rangle)}{\Gamma_p(\langle \frac{h p^i}{d} \rangle)} = \phi(b).$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \|a\|_{\#} + \|x^{\natural} - a\|_{\#} &\geq \|x^{\natural}\|_{\#} \\ &\geq \|x^{\star}\|_{\#} \\ &= \|x^{\natural} + (x^{\star} - x^{\natural})\|_{\#} \\ &= \|a + (x^{\natural} - a) + z_1 + z_2\|_{\#} \\ &\geq \|a + z_1\|_{\#} - \|(x^{\natural} - a) + z_2\|_{\#} \\ &\geq \|a + z_1\|_{\#} - \|x^{\natural} - a\|_{\#} - \|z_2\|_{\#} \\ &= \|a\|_{\#} + \|z_1\|_{\#} - \|x^{\natural} - a\|_{\#} - \|z_2\|_{\#}. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{l} \inf_{x_m} F_{\tau_m}^{k-1}(x_{[n]}, x_m, \Psi_m) := f_{\tau_m}(x_{[n]}, x_m, \Psi_m) + \mathcal{Q}_m^{k-1}(x_{[n]}, x_m) \\ g_{\tau_m}(x_0, x_{[n]}, x_m, \Psi_m) \leq 0, \\ [A_{0,m}, \dots, A_{\tau_m,m}][x_0; x_{[n]}; x_m] = b_m \\ x_m \in \mathcal{X}_{\tau_m}. \end{array} \right.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\gamma^0 = \begin{pmatrix} \sigma^3 & 0 \\ 0 & -\sigma^3 \end{pmatrix}, \gamma^1 = \begin{pmatrix} i\sigma^2 & 0 \\ 0 & -i\sigma^2 \end{pmatrix}, \gamma^2 = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & i\sigma^1 \end{pmatrix}, \gamma^3 = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$g^{\dagger} = \begin{pmatrix} 1 + \frac{1}{4}\eta\eta^{\diamond} & \frac{1}{2}(b\eta^{\diamond} + a^{\diamond}\eta) & -\frac{1}{2}(a\eta^{\diamond} - b^{\diamond}\eta) \\ \frac{1}{2}\eta^{\diamond} & a^{\diamond}(1 - \frac{1}{8}\eta\eta^{\diamond}) & b^{\diamond}(1 - \frac{1}{8}\eta\eta^{\diamond}) \\ \frac{1}{2}\eta & -b(1 - \frac{1}{8}\eta\eta^{\diamond}) & a(1 - \frac{1}{8}\eta\eta^{\diamond}) \end{pmatrix}$$